

Why ESP32 DMX receive works, and the only way it breaks: the interrupt-latency budget

DMX in: 1 byte every 44 μ s ($\sim 22,700$ bytes/s, continuous, no native framing)

UART RX FIFO = 128 bytes = ~ 5.6 ms of slack

overflow:
slots dropped

typical ISR latency
 $\sim$ tens of μ s (safe)

WiFi MAC burst
 $\sim$ hundreds of μ s (safe)

NON-IRAM ISR during a flash write,
long noInterrupts(), or CPU starvation
 > 5.6 ms $\rightarrow$ DROPPED FRAME

esp_dmx keeps its receive ISR in IRAM, so flash-cache blackouts do not stall it; pin the DMX work to a core away from WiFi and never block interrupts longer than the FIFO depth.

0

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8

time the receive ISR is delayed (ms)